

Simple Laws, the Arrow of Time, Many Worlds, Superdeterminism and All: Convergence from Mathematical Monism

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Abstract

The Simplicity Dominance Principle (SDP) states that a mathematical structure's measure is governed by its algorithmic probability: $P(S) \propto 2^{-K(S)}$, where $K(S)$ is the Kolmogorov complexity of its shortest generating description. SDP follows from the thesis that reality is identical to mathematical structure (Bernstein, 2026c) and that algorithmic probability is the natural measure over structure-space (Schmidhuber, 2000; Bernstein, 2026d).

Five results follow from SDP: (1) Occam's Razor is a theorem about the statistics of existence; (2) SDP selects for universes whose physics takes variational form, grounding the Principle of Least Action as a selection effect across structure-space; (3) cosmological fine-tuning dissolves; (4) the thermodynamic arrow of time is grounded rather than postulated; (5) superdeterminism is rehabilitated as a serious interpretation of quantum mechanics. A sixth result, the arrow of computation, provides the structural foundation for the time asymmetry SDP quantifies: the inverse of $1+1=2$ does not exist, yielding an irreversibility that holds without reference to measure or probability. SDP's time arrow depends on this irreversibility: end states have high K precisely because many-to-one functions ran forward. Under mathematical monism this structural fact is as fundamental as unitarity. Together with SDP, it strengthens the Many-Worlds Interpretation: MWI is the only framework where the substrate stays reversible while the arrow exists at the function level. Further consequences extend beyond physics and are developed in companion papers. That convergent results span epistemology, dynamics, cosmology, thermodynamics, computation, quantum foundations, variational dynamics, and variational geometry constitutes cumulative evidence for the underlying framework.

Keywords: Simplicity Dominance Principle, Kolmogorov complexity, algorithmic probability, Occam's Razor, arrow of time, superdeterminism, Many-Worlds, variational dynamics, variational geometry, mathematical monism

1. One Principle

Among all consistent mathematical structures, simpler ones are exponentially more probable. In a prefix-free coding, each program of length k occupies a fraction 2^{-k} of program-space. A structure producible by a 10-bit program has

$2^{10} = 1024$ times the measure of one requiring a 20-bit program. Each additional bit of specification halves the fraction. This is the Simplicity Dominance Principle. A “program” here is a finite description, not a process requiring execution. The structures exist as mathematics. The measure is over specifications, not computations.

The philosophical foundation is developed elsewhere: reality is identical to mathematical structure because the pattern-randomness dichotomy exhausts logical space (Bernstein, 2026c), and algorithmic probability is the unique natural measure over structure-space (Schmidhuber, 2000; Bernstein, 2026d). This paper does not re-derive the foundation. It demonstrates what SDP explains.

The argument for SDP is cumulative. Any single consequence might be coincidence. Six independent consequences from the same principle, across unrelated domains, is the kind of convergence that distinguishes a correct principle from an ad hoc hypothesis.

2. Occam’s Razor Is a Theorem

Scientists prefer simpler theories. This preference has been treated as methodological taste, aesthetic bias, or pragmatic convenience. Hossenfelder (2018) asked why physics should prefer simplicity at all, and found no satisfactory answer. SDP provides one: lower-K structures have exponentially higher measure. The criterion is not beauty but brevity.

A theory with generating complexity K describes structures with measure proportional to $2^{-(K)}$. A competitor with complexity $K' > K$ describes structures with measure proportional to $2^{-(K')}$. The ratio is $2^{(K - K')}$, exponentially favoring the simpler theory. Occam’s Razor is not a guide to inference. It is a fact about the distribution of existence.

This explains why simplicity has been so unreasonably effective as a criterion. Every major theoretical unification in physics (Maxwell unifying electricity and magnetism, Einstein unifying space and time, the Standard Model unifying three forces) reduced the total descriptive complexity of physics. Each was rewarded by the universe confirming its predictions. SDP says this pattern will continue: observers are exponentially more likely to find themselves in structures with lower generating complexity, so the final theory of physics should be simpler than what we have, not more complex.

3. The Principle of Least Action

Classical mechanics, general relativity, quantum field theory, and the Standard Model all admit variational formulations. Systems evolve along paths extremizing the action integral. Why should nature prefer extremal paths?

Consider two ways to specify the same physics. Specification A: “systems follow paths extremizing the action functional S .” Specification B: exhaustive enu-

meration of permitted trajectories. Specification A has vastly lower generative complexity. Under SDP, variational universes dominate the measure. SDP does not make PoLA emerge within a universe. It selects for universes whose physics is variational, because variational specifications have lower generative complexity. The Principle of Least Action is a selection effect across structure-space, not a dynamical consequence within any particular structure. A deeper result connects PoLA to the computational arrow: both are instances of many-to-one mappings. Many paths, one classical survivor. Many inputs, one output. The Noether chain (observers $\rightarrow$ stability $\rightarrow$ conservation $\rightarrow$ symmetry $\rightarrow$ variational) forces variational physics for any observer-supporting universe. Full development is given in Bernstein (2026k).

This connects to Noether’s theorem: every continuous symmetry of the action yields a conservation law. Symmetries reduce the independent specifications needed. More symmetry means lower K. SDP predicts that fundamental physics will be maximally symmetric, which is what we observe: the Standard Model is built on gauge symmetries, and general relativity on diffeomorphism invariance.

4. Fine-Tuning Dissolves

Change the gravitational constant slightly and stars cannot form. Adjust the strong force and atoms fall apart. The universe appears calibrated for complex life.

If all consistent mathematical structures exist (which follows from the null filter argument: any filter is itself a consistent structure, so all filters exist, including none), then observer-permitting structures necessarily exist among them. We find ourselves in one because we could not find ourselves anywhere else. Selection bias, not design.

SDP adds precision: among observer-permitting structures, simpler ones are exponentially more probable. This predicts that the fundamental constants are not arbitrary but among the simplest values compatible with observers. It also predicts that outstanding puzzles like dark matter will have simple solutions (lower-K explanations will be confirmed over higher-K alternatives), and that the initial conditions we observe are low in complexity because observers are more likely to find themselves computing from simple starting points. The same bias applies to temporal location: earlier observer-moments have shorter specification prefixes, but later moments have more total observers. The distribution peaks where these pressures meet. SDP also applies to competing physical theories: string theory’s total specification complexity (approximately 3,000 to 4,700 bits) dwarfs any plausible generating rule for the Standard Model’s 19 to 26 scalar parameters, rendering string theory exponentially disfavored (Bernstein, 2026g).

5. The Arrow of Time

The laws of physics are time-symmetric. Yet entropy increases. The standard approach (Albert, 2000) postulates a low-entropy initial condition, the Past Hypothesis, as a brute boundary condition. SDP derives it.

A universe described forward from initial conditions I has total complexity $K(L) + K(I)$. If I is low-entropy (smooth, nearly uniform), $K(I)$ is low. The same universe described backward from final state I_f has complexity $K(L) + K(I_f)$, where I_f encodes the accumulated output of billions of years of dynamical evolution, enormous and potentially incompressible.

Forward-described structures dominate the measure by $2^{(K(I_f) - K(I))}$, an astronomical factor. All states exist eternally in the block universe. Observers experience the low-complexity end as the past because computation is many-to-one and consciousness can only model forward. Low entropy is low descriptive complexity. SDP exponentially favors low complexity. The Past Hypothesis is not a brute postulate about cosmology but a consequence of what modeling means. Full development is given in Bernstein (2026e, 2026h).

The computational arrow provides the structural foundation for this asymmetry. Nearly all computations are many-to-one: $1+1=2$ is deterministic, but $2=x+y$ is not. The inverse does not exist. This is why $K(\text{end state})$ is high: many-to-one functions ran forward, and any backward walk would require specifying which of the many inputs produced each output. SDP quantifies the measure-theoretic consequence of this irreversibility. The computational arrow stands without SDP. SDP's time arrow depends on it. Deutsch's constructor theory supports the point: it defines physics through local operations, not global state descriptions. When Deutsch appeals to global unitarity to dissolve the arrow, he leaves his own framework. The arrow lives at the level constructor theory selects (Bernstein, 2026h).

6. Superdeterminism Rehabilitated

Bell's theorem shows that no local hidden-variable theory satisfying statistical independence (SI) can reproduce quantum predictions. The mainstream dismisses superdeterminism (rejecting SI) as requiring "implausible fine-tuning" of initial conditions.

This objection conflates specificity with complexity (Bernstein, 2026b). Initial conditions generated by simple rules have low K and are measure-favored, regardless of how specific they appear. The digits of π are specific but not complex: a short program generates all of them.

The K -decomposition clarifies the comparison. Total specification complexity is $K(L) + K(I) + K(O)$, where $K(L)$ is laws, $K(I)$ is initial conditions, and $K(O)$ is the specification of which particular measurement outcomes occurred. Under collapse interpretations, $K(O)$ grows without bound: each measurement selects one eigenvalue, accumulating an algorithmically incompressible string. Under

both Many-Worlds and superdeterminism, $K(O) = 0$. No random selection occurs.

The fine-tuning objection inadvertently argues against collapse, not against superdeterminism. Many-Worlds retains a parsimony advantage with current models (the Schrödinger equation alone, no additions), but the competition is mathematical: whichever generating rule has lower total K wins.

7. Beyond Physics

The framework extends beyond physics: consciousness, AI alignment, and ethics follow as structural consequences and are developed in companion papers (Bernstein, 2026f, 2026i).

8. Convergence

Five results from SDP, one independent structural fact, two from the combination:

1. Occam's Razor (epistemology)
2. Least Action (selection across structure-space)
3. Fine-tuning (cosmology)
4. Arrow of time (thermodynamics)
5. Superdeterminism (quantum foundations)
6. Arrow of computation (structural foundation: many-to-one functions)
7. Many-Worlds strengthened (SDP + computational arrow)
8. Variational dynamics forced (many-to-one + Noether chain, Bernstein 2026k)
9. Variational geometry forced (metric bundle as emergent continuum form with harmonic overtones above, Bernstein 2026n)
10. String theory's vibrational spectrum redeemed as derived geometric overtones rather than postulated objects (Bernstein 2026n)
11. Encoding hierarchy: Turing mapping preserves data, the holographic principle preserves physics within entropy bounds, jet bundle reconstruction preserves geometry exactly: three distinct senses of encoding suggesting a deeper structural pattern (Bernstein 2026n, 2026o)

Each was previously treated as a separate puzzle. Mathematical monism unifies them. The computational arrow is the structural foundation. SDP quantifies its consequence. Together they converge on a single framework.

The principle makes a meta-prediction: future puzzles in fundamental physics will have simple solutions. Complexity is the exception, not the rule. If the pattern holds, SDP is confirmed. If a future discovery requires irreducibly high- K explanation, SDP is challenged. The principle is falsifiable through its consequences.

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